Celebal Excellence Internship Program 2026
Mini Project Report

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Week: 4
Domain: Data Engineering

Objective

The objective of this mini project was to learn the basics of Microsoft Azure Cloud and build a complete data pipeline using Azure Data Factory (ADF). The project involved storing a CSV file in Azure Blob Storage, creating a pipeline in Azure Data Factory, validating the file metadata, copying the file to another location, and monitoring the pipeline execution.

Problem Statement

Build a complete data pipeline that reads a CSV file from Azure Blob Storage and processes it using Azure Data Factory.

Tools and Services Used

- Microsoft Azure Portal
- Azure Resource Group
- Azure Storage Account
- Azure Blob Storage
- Azure Data Factory (ADF)
- Azure IAM (Identity and Access Management)
- Superstore CSV Dataset

Step 1: Create Azure Resources

Description

First, a Resource Group was created to manage all Azure resources. Then, a Storage Account was created for storing files.

Steps Performed

- Logged in to Azure Portal.
- Created a Resource Group.
- Created an Azure Storage Account.
- Created two Blob Containers:
  - **source-data**
  - **destination-data**
- Uploaded the **Superstore.csv** file into the **source-data** container.

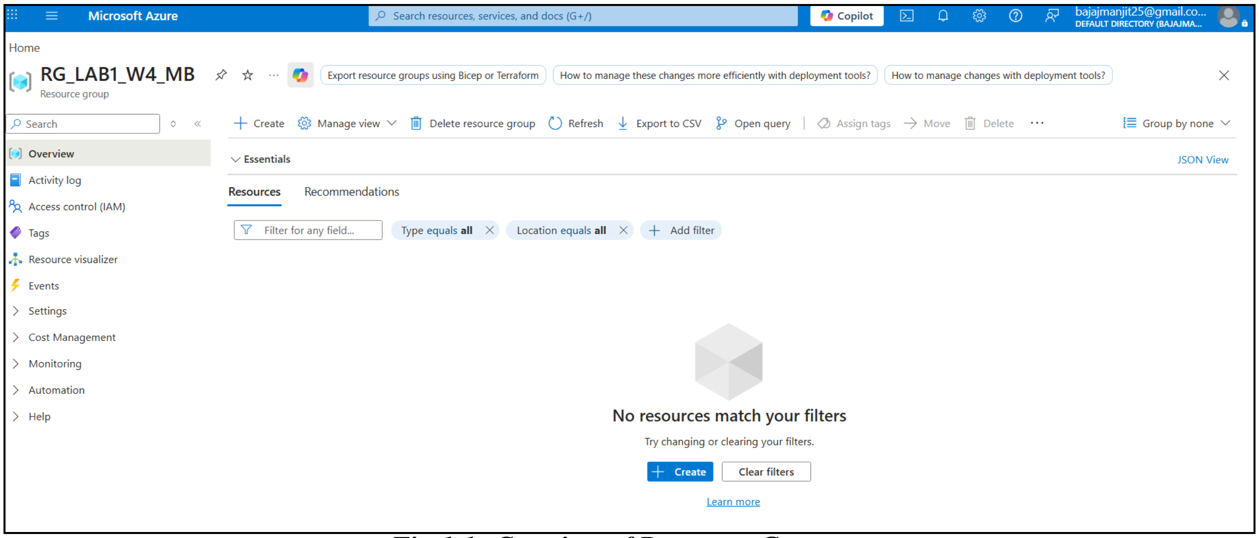

Fig 1.1: Creation of Resource Group

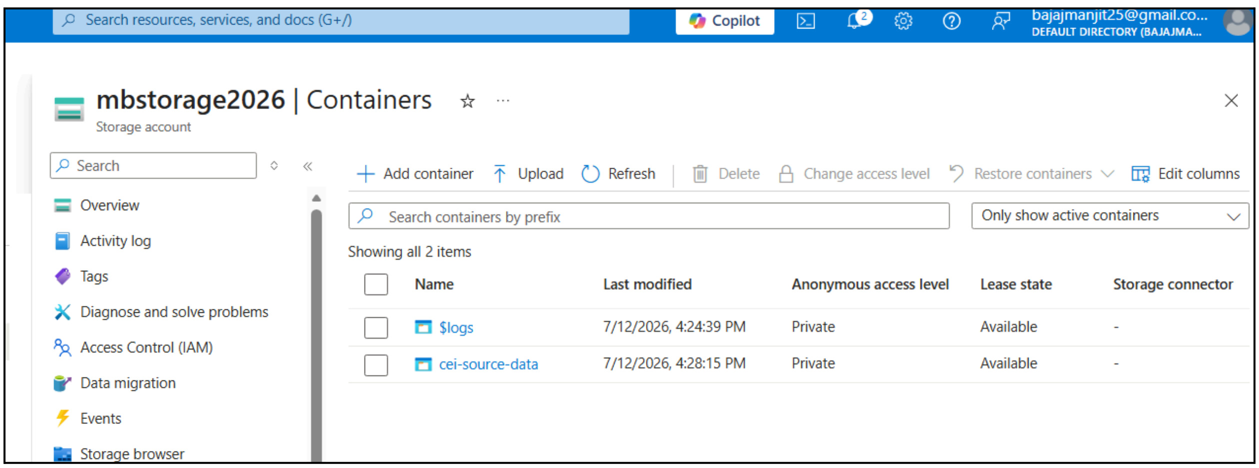

Fig 1.2: Creation of Blob Container

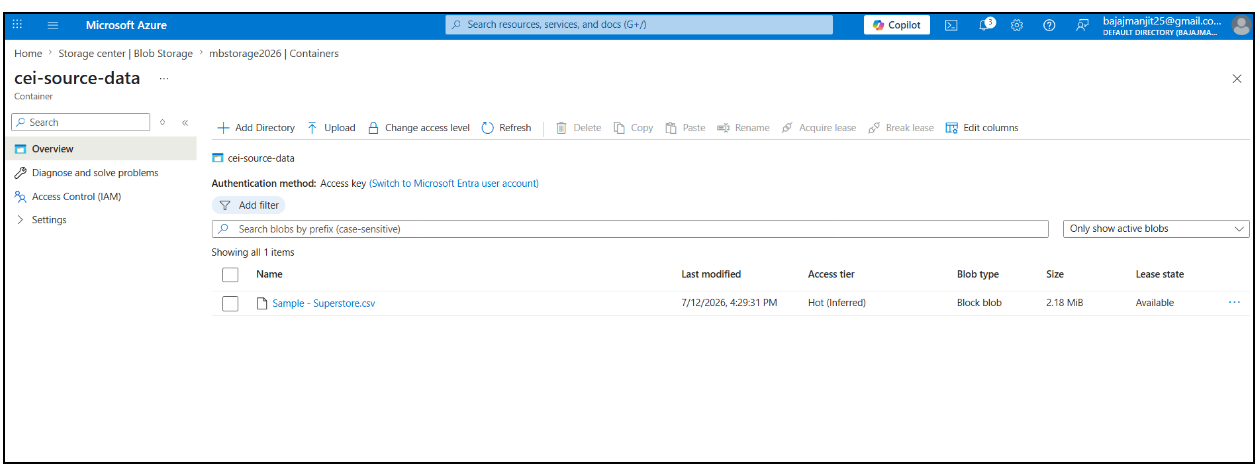

Fig 1.3: Upload superstore dataset csv file

Step 2: Create Azure Data Factory

Description

An Azure Data Factory instance was created to design and manage the data pipeline.

Steps Performed

- Created Azure Data Factory.
- Opened Azure Data Factory Studio.
- Explored the following sections:
  - Author
  - Monitor
  - Manage

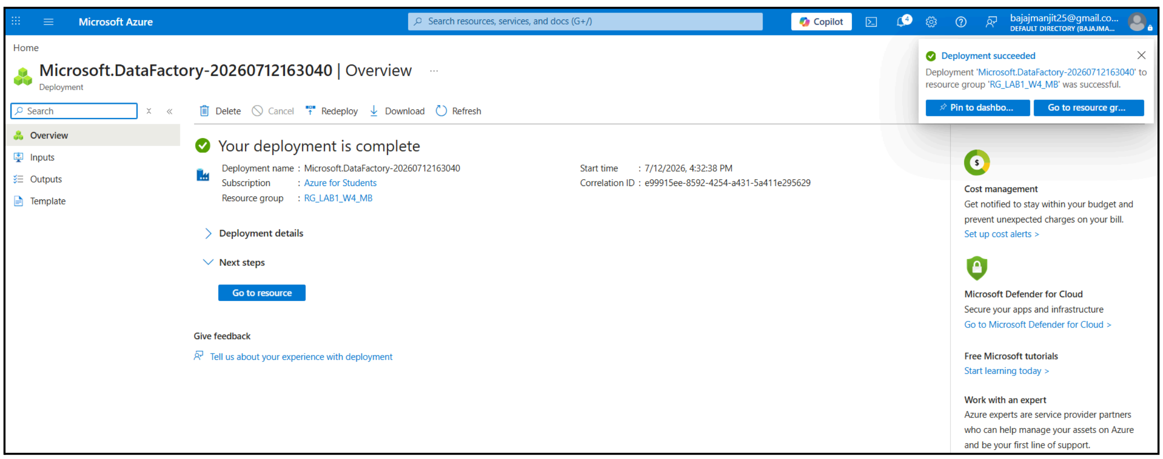

Fig 2.1: Creation of Azure Data factory

Step 3: Configure Linked Service and Datasets

Description

A Linked Service was created to connect Azure Data Factory with Azure Blob Storage. Source and destination datasets were then configured.

Steps Performed

- Created a Linked Service for Azure Blob Storage.
- Created the Source Dataset pointing to:
  - source-data
  - Superstore.csv
- Created the Destination Dataset pointing to:
  - destination-data
  - Superstore_Copy.csv

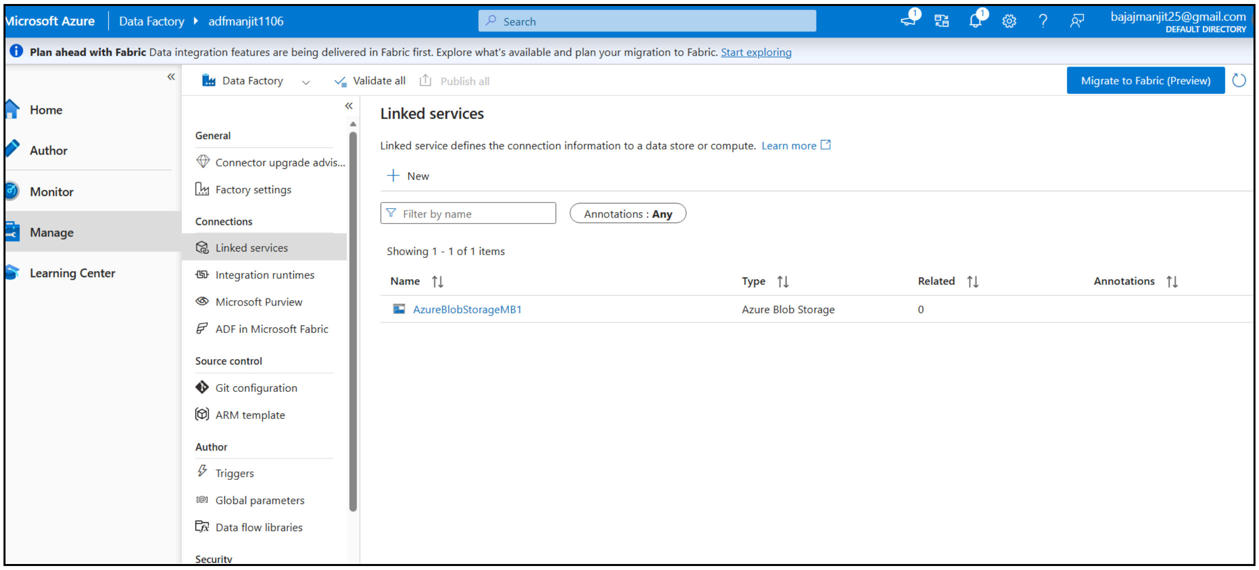

Fig 3.1 Create Linked Services

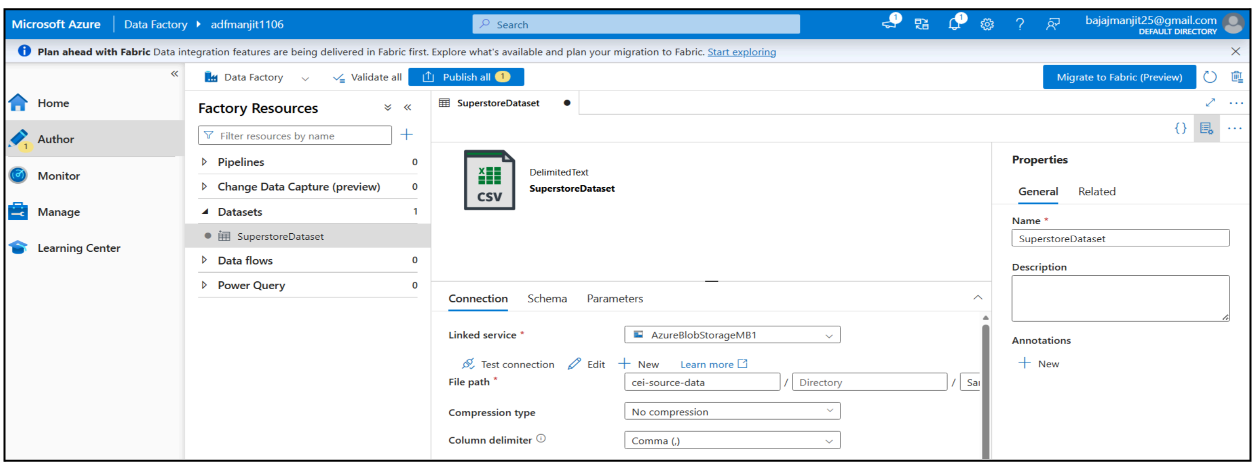

Fig 3.2: Created Source data point

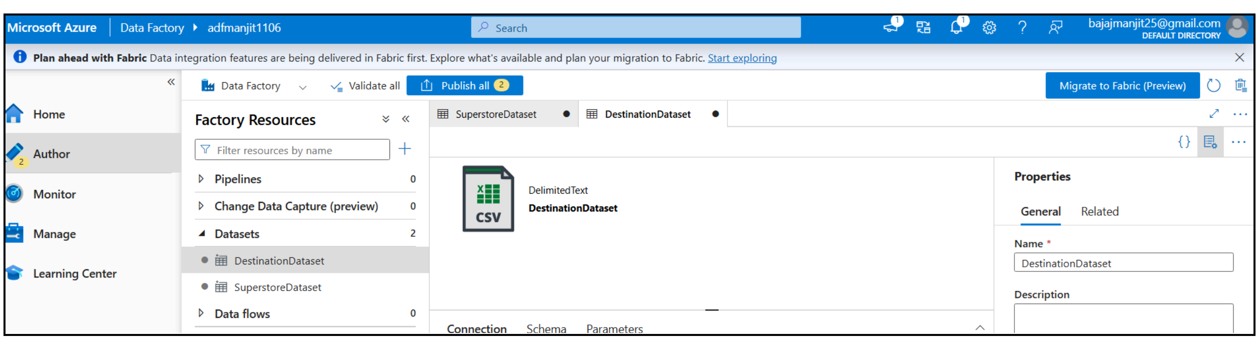

Fig 3.3: Created Destination data point

Step 4: Create Data Pipeline

Description

A pipeline was created to validate the file metadata and copy the file from the source container to the destination container.

Steps Performed

- Created a new pipeline.
- Added the **Get Metadata** activity.
- Configured it to retrieve:
  - File Exists
  - File Size
  - Last Modified Date
  - File Name
- Added the **Copy Data** activity.
- Connected **Get Metadata** to **Copy Data**.
- Configured the Source and Destination datasets.

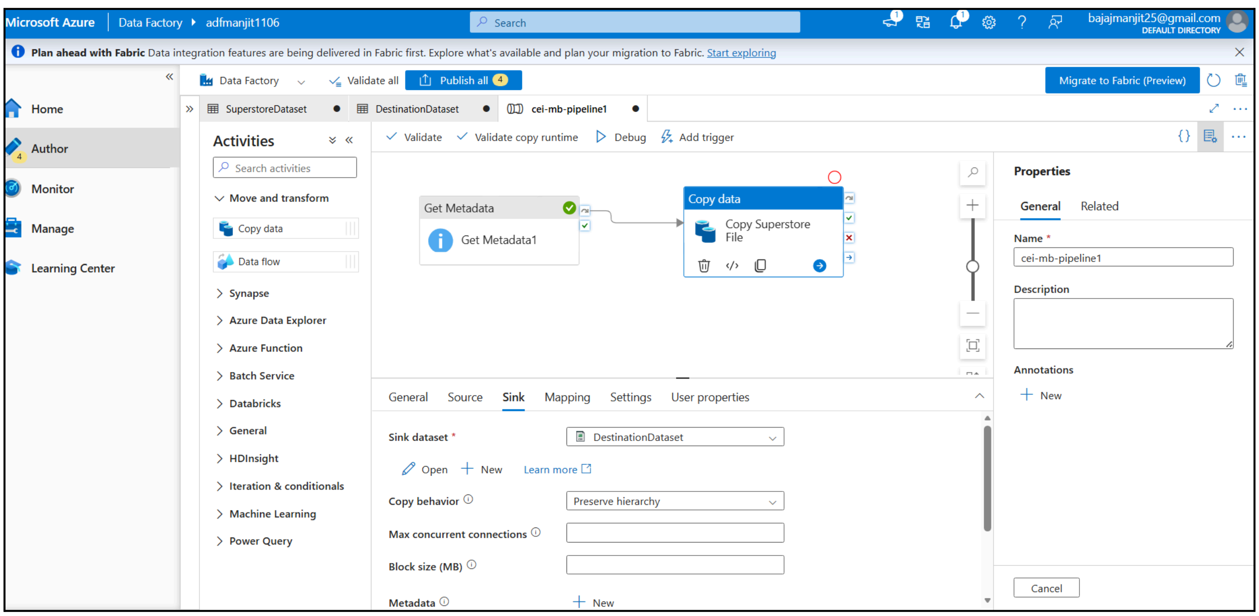

Fig 4.1: Creation of Data pipeline

Step 5: Execute the Pipeline

Description

The pipeline was executed using the Debug option. After execution, the pipeline status was checked in the Monitor section.

Steps Performed

- Validated the pipeline.
- Clicked **Debug**.

- Waited for execution.
- Verified that both activities completed successfully.

Result

The pipeline execution completed successfully with **Succeeded** status.

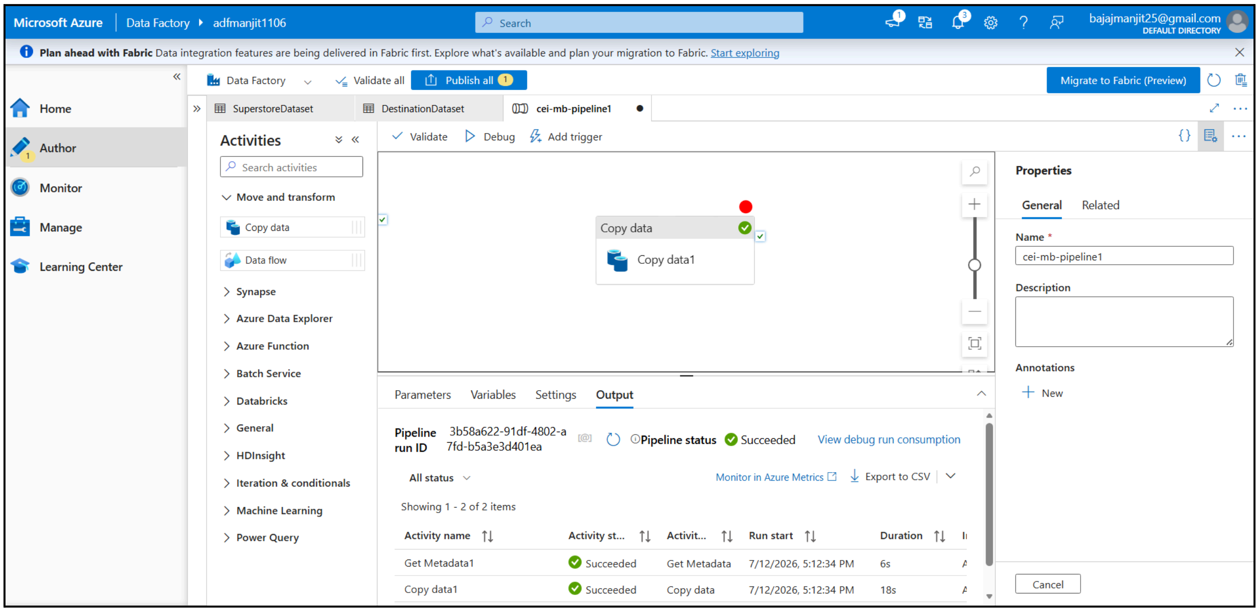

Fig 5.1: Execution of pipeline

Step 6: Metadata Validation

Description

The **Get Metadata** activity was used to verify the details of the source file before copying it.

Metadata Retrieved

- File Exists
- File Name
- File Size
- Last Modified Date

This confirmed that the correct source file was available before executing the copy operation.

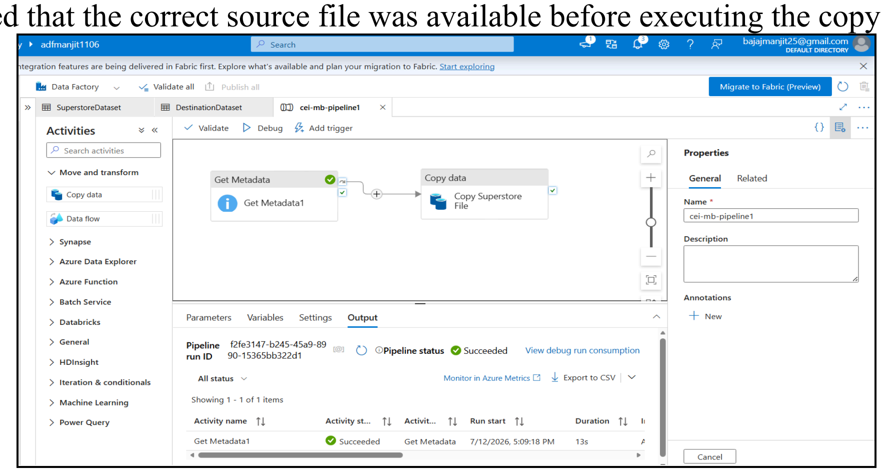

Fig 6.1: Successful Validation of Metadata

Step 7: Verify Copied File

Description

After the successful execution of the pipeline, the copied file was verified in the destination Blob container.

Result

The file **Superstore_Copy.csv** was successfully copied to the **destination-data** container.

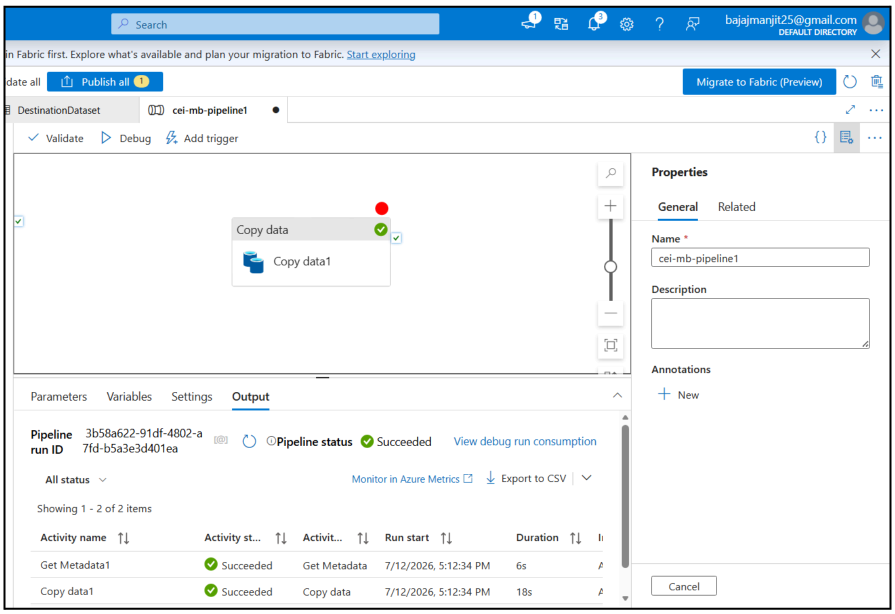

Fig 7.1: Successful Verification

Step 8: IAM Role Assignment

Description

Role-Based Access Control (RBAC) was configured to allow Azure Data Factory to securely access Azure Storage.

Roles Assigned

- Reader
- Contributor

The Azure Data Factory Managed Identity was granted the required permissions on the Storage Account.

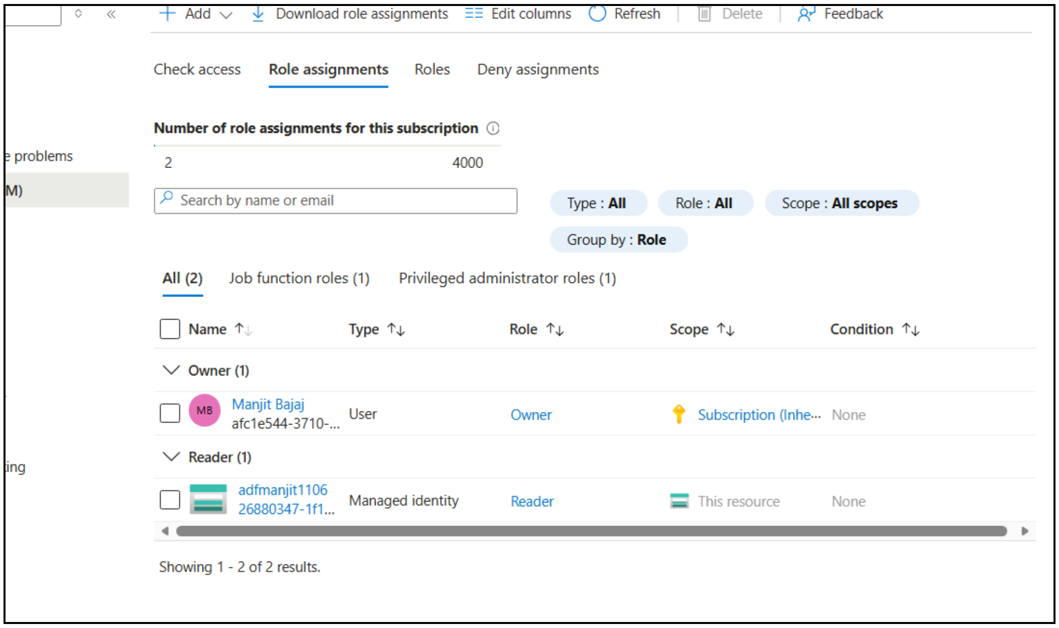

Fig 8.1: Role Assignment Succesful

Learning Outcomes

During this mini project, I learned how to use Microsoft Azure services to build a complete data pipeline. I gained hands-on experience in creating Azure resources, uploading files to Blob Storage, connecting Azure Data Factory with Storage using Linked Services, creating datasets, using the Get Metadata and Copy Data activities, executing and monitoring pipelines, and configuring IAM roles for secure access. This project helped me understand the basic workflow of Azure Data Factory and cloud-based data integration.

Conclusion

This mini project successfully demonstrated an end-to-end data pipeline using Microsoft Azure and Azure Data Factory. The Superstore CSV file was stored in Azure Blob Storage, its metadata was validated using the Get Metadata activity, and it was copied to a destination container using the Copy Data activity. The pipeline executed successfully, and the copied file was verified in the destination location. IAM roles were also configured to provide secure access between Azure Data Factory and Azure Storage. Overall, this project strengthened my understanding of Azure cloud services and provided practical experience in building and managing cloud-based data pipelines.